

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



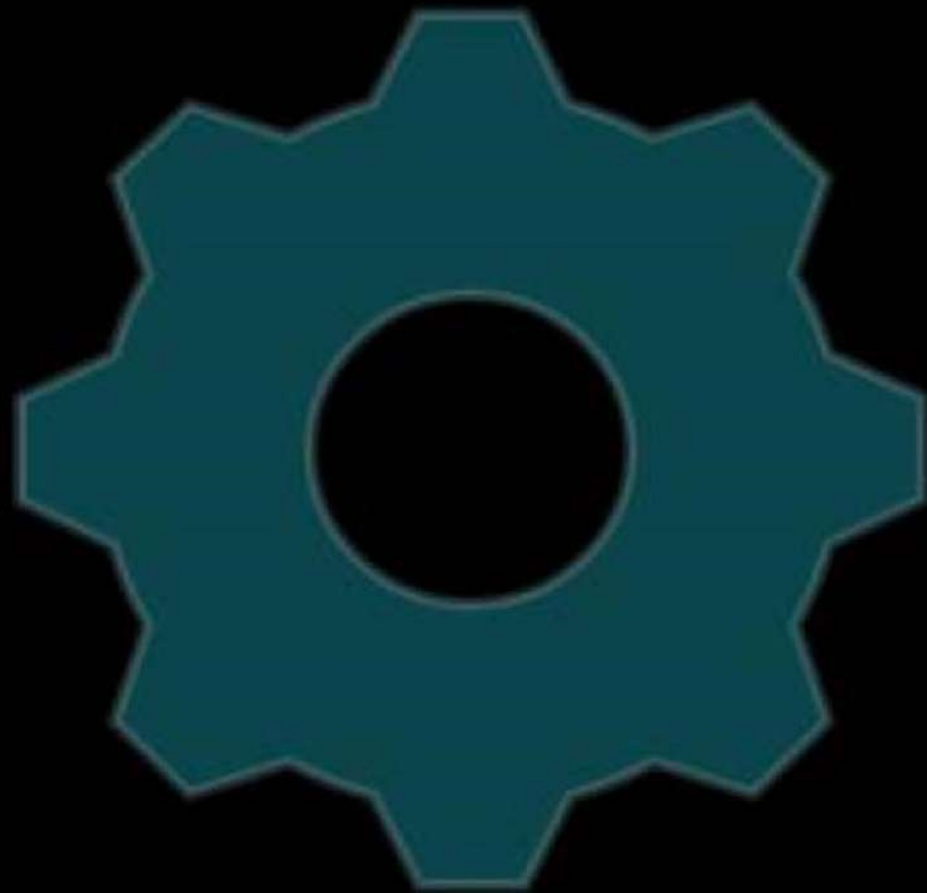
Lecture No.

2



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Topics to be Covered



o1  Combination With Repetition

o2 

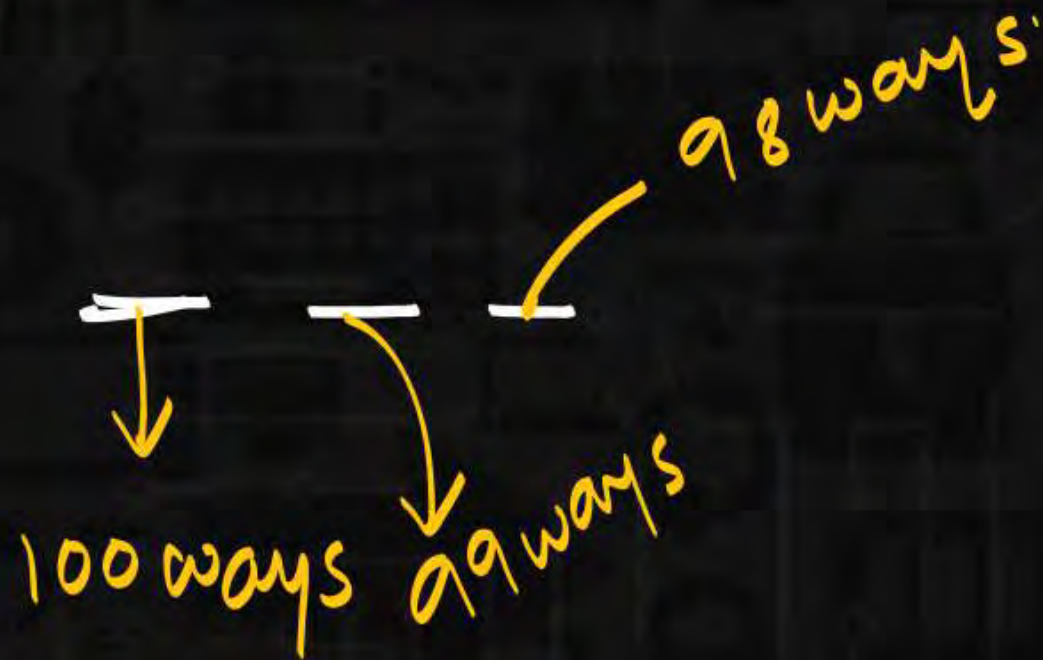
o3 

o4 

COMBINATORICS

$$n = 100$$

100 prizes, how many ways
we can distribute 3 prizes
among 3 students?



$$\begin{array}{c} 1 \quad k=3 \quad 2 \quad 3 \quad k \\ \downarrow \quad \downarrow \quad \downarrow \\ (n-0) \quad (n-1) \quad (n-k+1) \\ 100-0 \quad 100-1 \quad (100-3+1) \\ = 100 \quad = 99 \quad (100-2) \\ \quad \quad \quad 98 \end{array}$$

$$= \underline{100} \cdot \underline{99} \cdot \underline{98}$$

COMBINATORICS



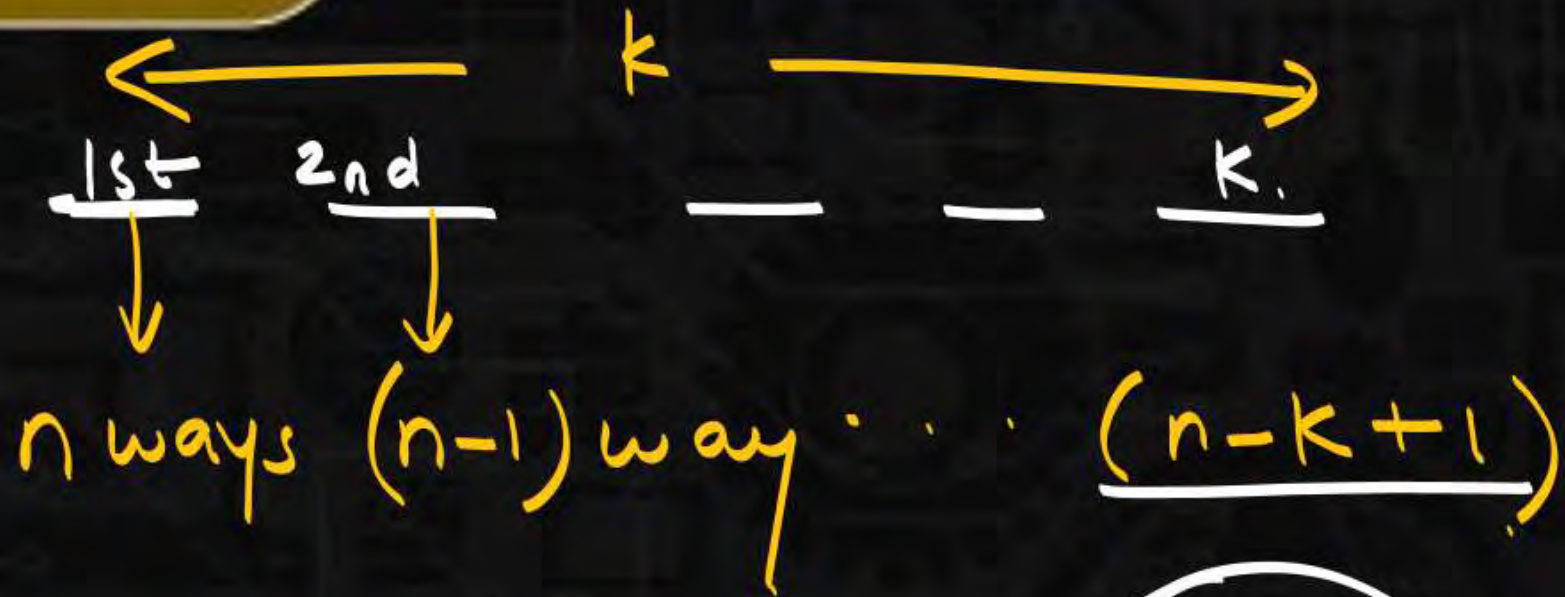
500
prizes

$$\begin{array}{c} \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \\ \downarrow \\ 500 \times 499 \times 498 \times 497 \end{array}$$

10000

$$\begin{array}{c} 0 \\ \hline \downarrow \\ 1000 \end{array} \quad \begin{array}{c} 9 \\ \hline 999 \end{array} \quad \begin{array}{c} 8 \\ \hline 998 \end{array} \quad \begin{array}{c} 7 \\ \hline 997 \end{array} \quad \begin{array}{c} 6 \\ \hline 996 \end{array}$$

COMBINATORICS



$$= n(n-1)(n-2)\dots(n-k+1) \frac{(n-k)!}{(n-k)!}$$
$$= \frac{n!}{(n-k)!} = {}^n P_k$$

COMBINATORICS

$\{a, b, c\}$

$${}^3P_2 = \frac{3!}{(3-2)!} = \frac{3 \cdot 2 \cdot 1}{1!}$$

how many ways
we can arrange 2 letters?

products

3×2
= 6 ways

visualization



Combination:

COMBINATORICS



Combinations:

larger set $\rightarrow$ smaller set

$\{a, b, c\}$
 $\uparrow \quad \downarrow \downarrow$

nCr

$\{a, b\}$

$\{b, c\}$

$\{a, c\}$

$r!$

Permutation:

ab
 ba

bc

cb

ac

ca

$$nCr = \frac{n!}{r! \times (n-r)!}$$

$$nCr \times r! = \frac{n!}{(n-r)!}$$

$$nCr \times r! = nPr$$

COMBINATORICS



Combination with Reptⁿ.

lady, she goes to mall, 3 containers.

Apple, orange, papaya.

how many ways she can
choose 4 fruits?



COMBINATORICS



A



O



P

manually:

4A
4O
4P

3

3A1O
3A1P
3O1A
3O1P
3P1A
3P1O

6

2A2O
2O2P
2P2A

3

2A1O1P
2O1A1P
2P1A1O

3

15 ways

✓✓✓✓
A A | O | P. Total
C line

A | O O | P. $6C_2$

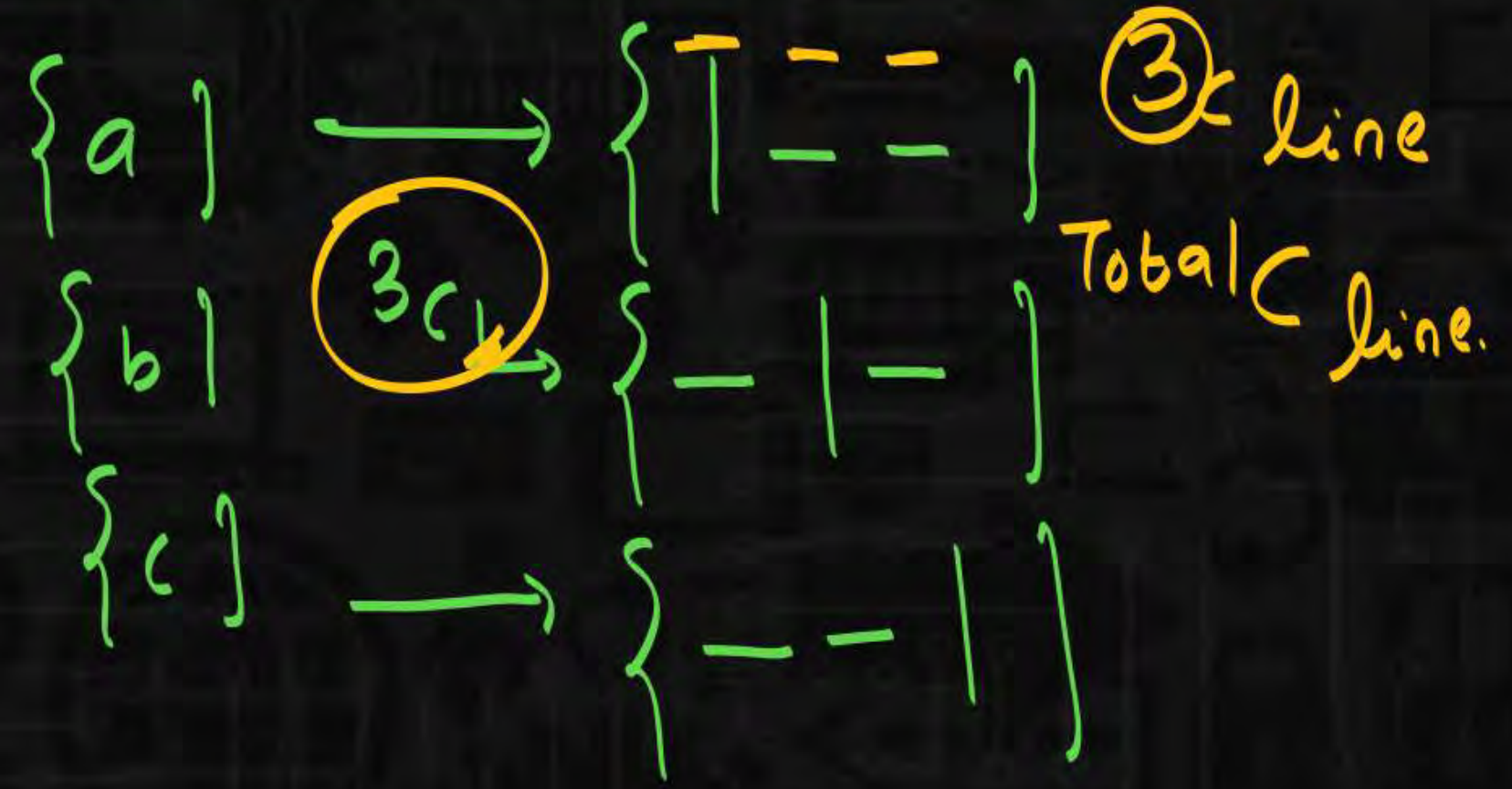
A | O O O |

$6C_2 = \frac{6 \cdot 5}{2 \cdot 1} = 15 \text{ ways}$

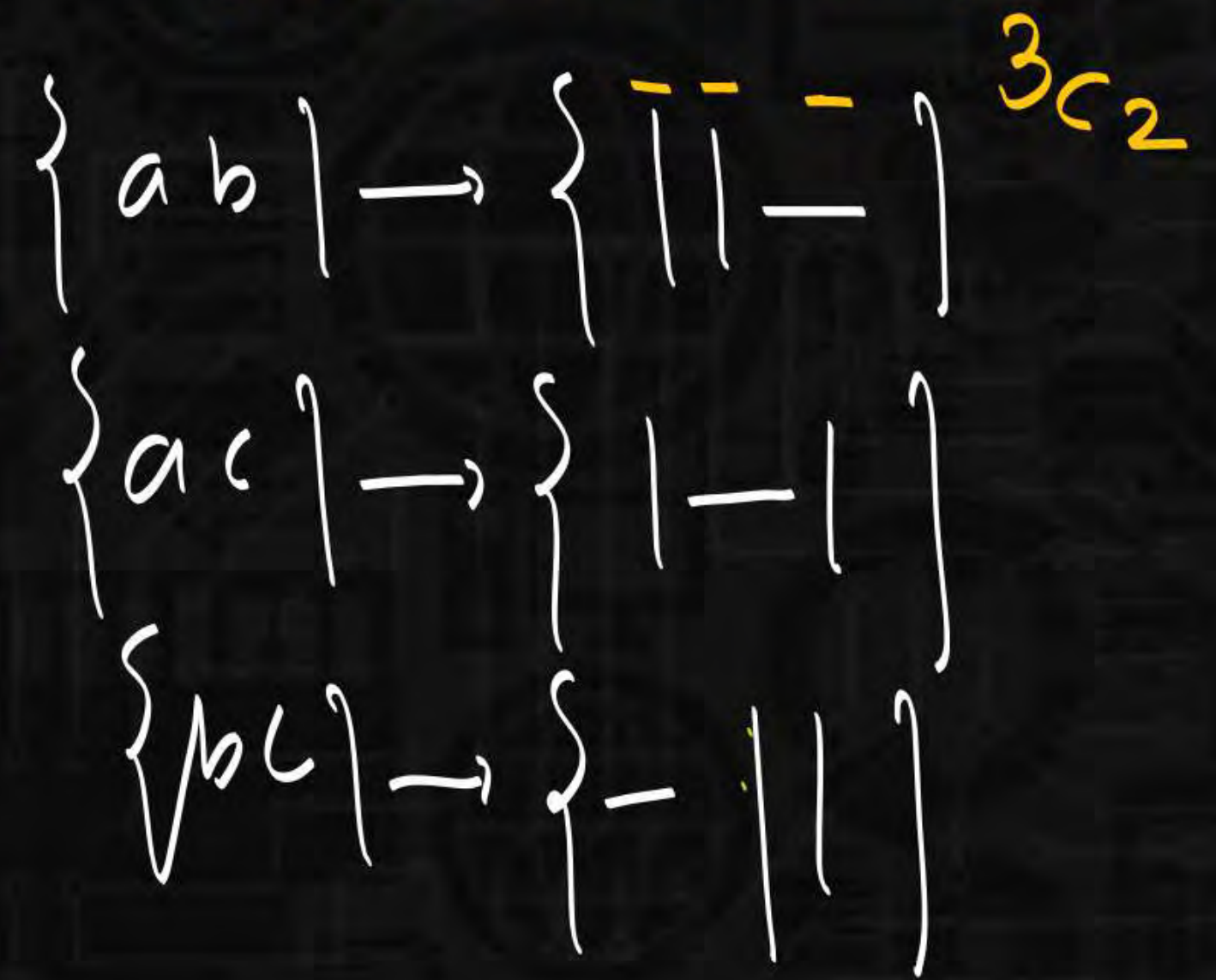
COMBINATORICS

$\{a, b, c\}$ selecting 1 element
= shifting 1 line

how many ways choose 1 element?



$\{a, b, c\} \rightarrow 3C_2$
how many ways we can
choose 2 elements?



COMBINATORICS



How many ways we can distribute
5 coins among 3 children?

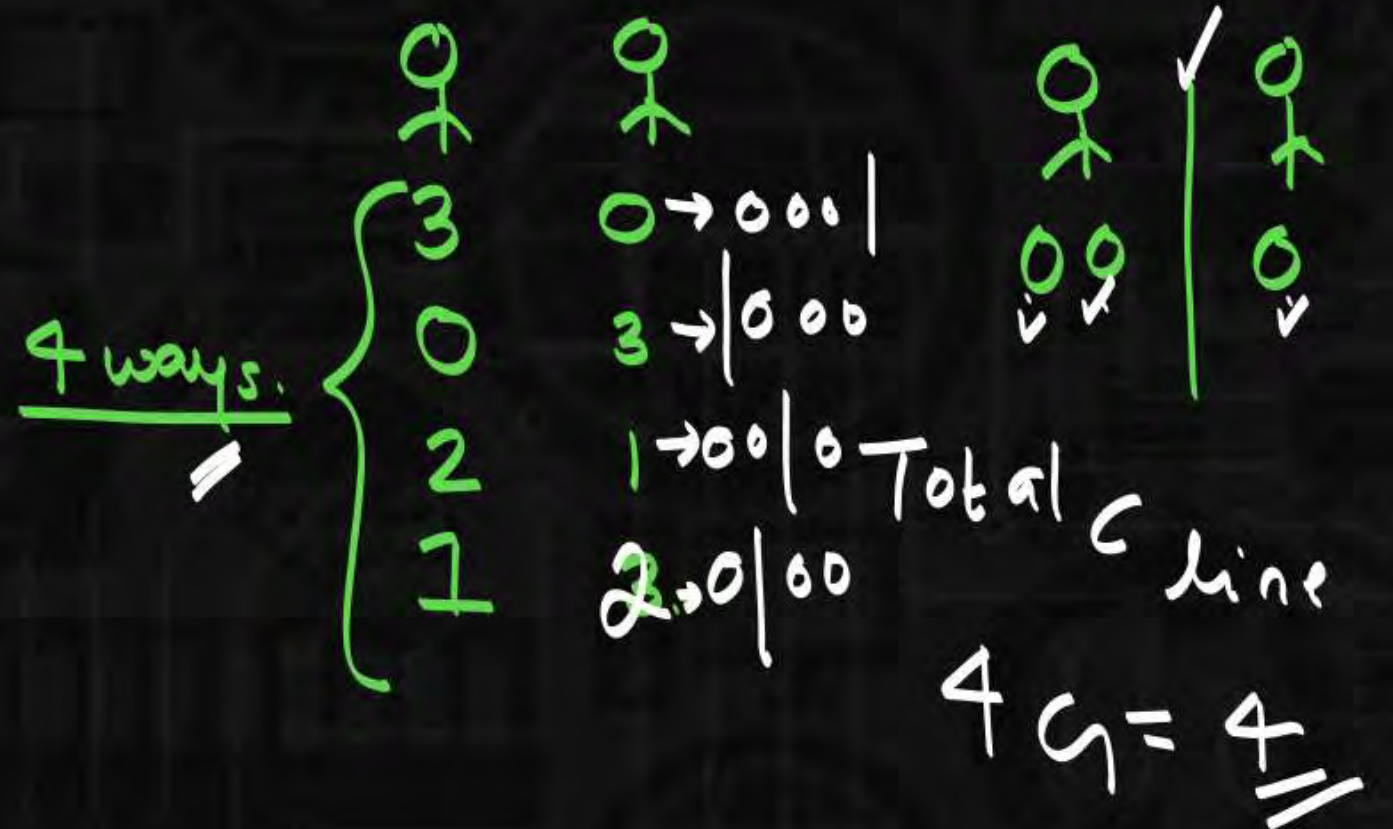
Similar



Total C line.

$$7C_2 = \underline{\quad}$$

how many ways we
can distribute 3 coins
among 2 children?

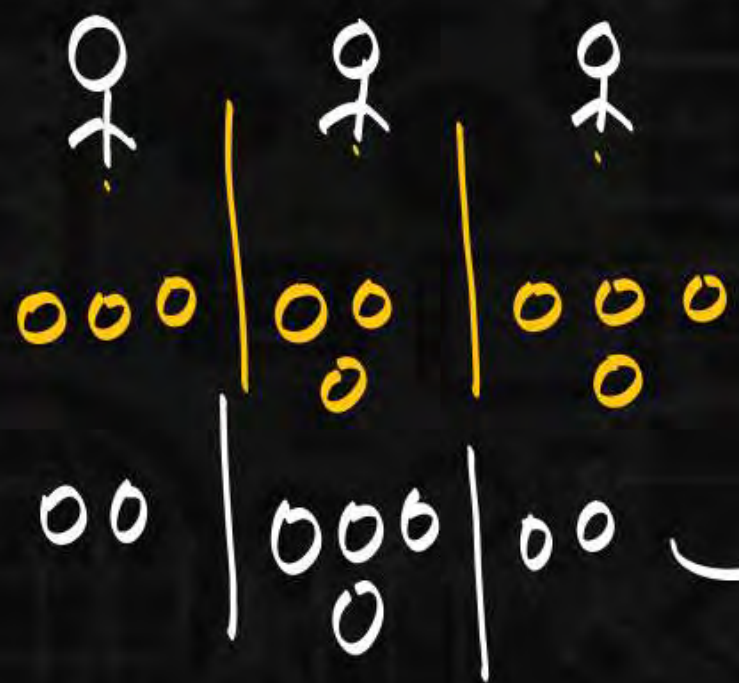


COMBINATORICS



how many ways we can
distribute 10 similar coins
among 3 students?

$x_1 + x_2 + x_3 = 10$ $x_i \geq 0$
how many non negative
integer solutions are
possible?



$$x_1 + x_2 + x_3 = 10$$

$$= 3 \quad = 3 \quad = 4$$

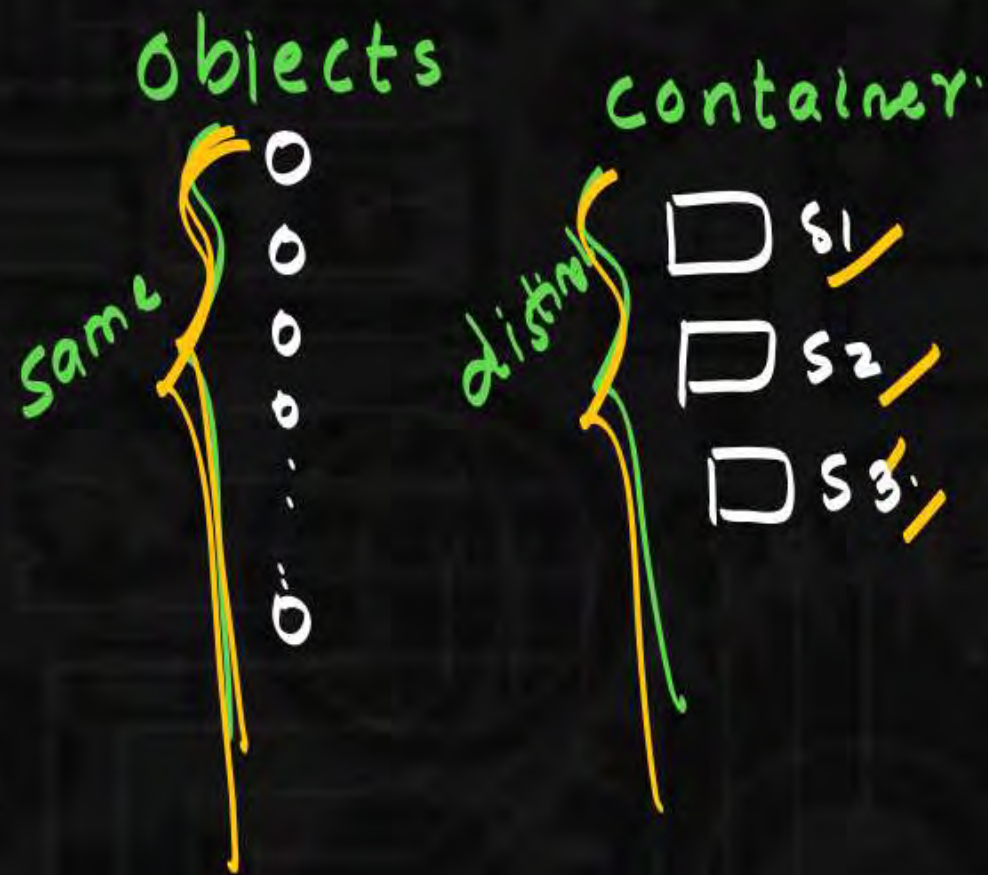
$$| 2C_2$$

$$x_1 = 2 \quad x_2 = 4 \quad x_3 = 2$$

COMBINATORICS



Objects containers
 10 similar coins $\rightarrow$ 3 students



s_1	s_2	s_3
10	0	0
object same	container diff.	→ maybe empty

C.R

Onto

Objects
 distinct

cannot empty

containers
 distinct

$\left\{ \begin{array}{c} a_1 \\ a_2 \\ a_3 \\ a_4 \\ \vdots \\ a_{10} \end{array} \right.$

$\left\{ \begin{array}{c} n_1 \\ n_2 \\ n_3 \end{array} \right.$

COMBINATORICS

How many ways we can distribute 10 similar coins among 3 students?

=

$$x_1 + x_2 + x_3 = 10 \quad x_i \geq 0$$

$$\begin{array}{c|c|c} \text{♂} & \text{♂} & \text{♂} \\ 10 & 0 & 0 \end{array} \quad \begin{array}{l} 10+2 \\ 12 \end{array} \begin{array}{l} C_2 \\ C_2 \end{array}$$

how many ways we can distribute

10 similar coins among 3 students

such that each student will get at least 1 coin?

COMBINATORICS



$$10 - 3 = 7$$

$$7 + 2 = 9 \Rightarrow 9C_2$$

$$x_1 + x_2 + x_3 = 10 \quad x_i \geq 1$$

how many ways we can distribute
10 similar coins among 3 students
such that each student will get atleast 1 coin?

COMBINATORICS



10 coins 3 strds.



$$10 + 2C_2 = 12C_2$$

$$10 - 3 = 7$$

7 coins



$$7 + 2C_2 = 9C_2$$

COMBINATORICS



$$x_1 + x_2 + x_3 = 10 \quad x_i \geq 1.$$

$$x_i \geq 1.$$

$$\frac{x_i - 1 \geq 0}{y_i}$$

$$y_i = x_i - 1$$

$$y_i + 1 = x_i$$

$$y_1 + 1 = x_1$$

$$y_2 + 1 = x_2$$

$$y_3 + 1 = x_3.$$

$$y_1 + 1 + y_2 + 1 + y_3 + 1 = 10$$

$$y_1 + y_2 + y_3 = 10 - 3 \quad y_i \geq 0.$$

$$y_1 + y_2 + y_3 = 7 \quad y_i \geq 0$$

$$7 + 2 + 2$$

COMBINATORICS



Q: how many ways we can distribute 10 similar coins among 3 students such that each student must get at least 2 coins?

=

Ans: $6C_2$.

$$x_1 + x_2 + x_3 = 10$$

$$x_i \geq 2.$$

COMBINATORICS



$$x_1 + x_2 + x_3 = 20$$

$$\rightarrow y_1 + 2 + y_2 + 3 + y_3 + 4 = 20$$

$$y_i \geq 0$$

$$x_1 \geq 2 \quad x_2 \geq 3 \quad x_3 \geq 4$$

$$y_1 + y_2 + y_3 = 20 - (2 + 3 + 4)$$

$$y_1 + y_2 + y_3 = 11$$

$$\frac{x_1 - 2 \geq 0}{y_1}$$

$$\frac{x_2 - 3 \geq 0}{y_2}$$

$$\frac{x_3 - 4 \geq 0}{y_3}$$

$$x_1 - 2 = y_1$$

$$x_2 - 3 = y_2$$

$$x_3 - 4 = y_3$$

$$x_1 = y_1 + 2$$

$$x_2 = 3 + y_2$$

$$x_3 = y_3 + 4$$

$$11 + 2C_2 = 13C_2$$

COMBINATORICS



HW

$$x_1 + x_2 + x_3 + x_4 = 32.$$

a) $x_i \geq 0$

b) $x_i \geq 0$ $x_i \geq 1$.

c) $x_1, x_2 \geq 5$ $x_3, x_4 \geq 7$

d) $x_i \geq 8$

e) $x_i \geq -2$

COMBINATORICS



$$x_1 + x_2 + x_3 + x_4 + x_5 = 15.$$

how many nonnegative solutions are possible if $x_1 + x_2 + x_3 = 6$.

$$= 6$$

$$= 9.$$

$$\boxed{x_1 + x_2 + x_3 + x_4 + x_5} = 15.$$

$$8C2$$

X

$$10C1$$

COMBINATORICS



(HW) $\left\{ \begin{array}{l} x_1 + x_2 + x_3 + \dots + x_7 = 37 \\ x_1 + x_2 + x_3 = 6 \end{array} \right.$

$5C3 \times 34C3$

$x_i > 0$

$x_i \geq 1$

(HW) how many ways we can toss 100 identical dice so that at least 3 of each face will be showing?

$87C92$

COMBINATORICS



m identical balls, n distinct bags. $m \geq kn$

how many ways can balls be placed each bag
contains at least k balls? (GATE 03)

$$\begin{array}{cccc} \binom{m-k}{n-1} & \binom{m-kn+n-1}{n-1} & \binom{m-1}{n-k} & \binom{m-kn+n-2}{n-k} \\ a & b & c & d \end{array}$$

COMBINATORICS



Total balls = m .

Total boxes = n . partitions
= $n-1$.

k

k

k

...

k

Total balls = m

Remaining balls = $m - nk$.

$$\frac{m - nk + n - 1}{n - 1}$$

